Joseph Wardle - Production TD

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Work Experience

Pipeline Technical Director - Student Short Films October 2025 - Present
Brigham Young University Center for Animation *Provo, UT*

- Develop **Python/Qt artist tools** and a **USD production pipeline** for **50+ artists** on two animated short films; troubleshoot material, lookdev, and publishing workflows in Maya and Houdini.
- Built a **Maya material-assignment visualizer** with colors chosen for colorblind accessibility, helping artists inspect assignments and identify unassigned objects on assets with many materials.
- Automated **material binding across Maya, Substance Painter, and Houdini**, mapping Maya shading groups through Substance texture sets to **USD geometry subsets**.
- Built a **Houdini material-library HDA** that generates **PxrSurface** shaders from published textures and paired **UsdPreviewSurface** shaders with JPEG proxies capped at 1K; automated color conversion and tiled EXR/TEX generation for RenderMan.
- Built a **Houdini lookdev HDA** with renderer-independent turntables, lighting presets, and artist-requested framing and angle controls; automated **Tractor rendering, denoising, and video export**.
- Revised material tools from **artist feedback** to preserve custom shader edits during rebuilds, support layered RenderMan materials, and simplify controls for specular ray depth, displacement, and subdivision.
- Extended **Houdini's Component Builder** with custom nodes for **geometry and material variants** and automatic viewport proxy generation.
- Established consistent **OCIO color workflows** across Houdini, Maya, Nuke, and Blender, adapting OpenDRT and ACES 2.0 transforms for OCIO 2.3 compatibility.

Web Developer March 2025 - November 2025
Brigham Young University - Computer Science Department *Provo, UT*

- Developed **Django services, REST APIs, and administrative interfaces** for nontechnical users.
- Built **GitHub Actions** workflows for testing, container builds, and deployment; used **Docker Compose** for local development and **Jira** for project tracking.

Selected Project

Cenote - GPU Path Tracer & Hydra Render Delegate github.com/joseph-wardle/cenote

- Built a **Rust/Vulkan GPU path tracer** with a **Hydra render delegate** integrating with usdview, husk, and Houdini.
- Implemented **OpenPBR shading in Slang**; validated energy conservation with **white furnace tests** and rendering results against **pbrt-v4**.

Education

Brigham Young University Provo, UT
B.S. Computer Science (Animation & Games Emphasis) *Expected May 2027*

GPA: 3.95. Coursework: Shading, Modeling, Rigging, FX, Linear Algebra, Data Structures, Multithreading.

Skills

Programming: Python, C++23, Rust, Slang, MEL, Bash | **UI:** Qt/PySide2, ImGui
Materials & Pipeline: USD, RenderMan, OCIO, Pixar Tractor, ShotGrid
DCCs: Maya, Houdini/Solaris, Substance Painter, Nuke, Blender
Development: Linux, Windows, Git, Perforce, GitHub Actions, Docker